

Report

Abstract

This project was aimed at developing and testing out a prediction-driven monthly stock trading strategy with a total of 50 NASDAQ stocks. We apply a rolling cross-sectional ridge regression model for 36 months, which will be trained on and forecast the next-month returns based on the information of the momentum, volatility, and market risk signals.

The final analysis made considerable advances from the primary concept. Our initial presentation, which focused on modeling the Top-20 universe, is switched to employing the full Top-50 universe, originally as proposed. It also utilizes the strict Rolling Window procedure rather than the Looking-Ahead methods.

Results (Out-of-Sample, Feb 2024 - Oct 2025):

Metric	Strategy	SPY
Ann. Return	35.75%	21.05%
Ann. Vol	32.20%	11.46%
Sharpe	1.05	1.66
Max DD	-16.90%	-
Hit Rate	50.00%	-

Conclusion: Evidently, the model successfully captured the characteristics of the 2021-2023 training period. The model showed strong effectiveness during the 2024-2025 recovery and AI-driven bull market. The stringent removal of look-ahead bias demonstrates that in these cases simple linear momentum factors demonstrate strong predictive power during the subsequent regime (e.g., sustained growth in semiconductor and tech sectors).

1. Introduction

1.1 Motivation

“With a provided universe of 50 liquid equities, can we predict future performance via a rolling linear model?”

Moving to a portfolio-wide focus and orientation of cross-sectional rankings is the main thrust of this project.

1.2 Research Questions

1. Can temporal (rolling) cross-sectional regression methods result in effective stock selection rankings?

2. Is there a benefit of making a Top 5 selection instead of just operating at the S&P 500 benchmark when taking risks into account?
3. What are the different ways in which market volatility levels impact strategy performance?

1.3 Methodological Rigor

One main objective of this last report is **Methodological Integrity**. Rather than naive train-test splits, we enforce:

At month t : Train on $[t - 36, t - 1]$ → Predict for t

The strategy is hence achieving real-world validation with no look-ahead bias.

2. Data Description

2.1 Stock Universe

We enlarged the universe from initial 20 stocks to the **Top 50 NASDAQ stocks** (as required by the Proposal) to capture higher cross-sectional dispersion.

Sectors: - Technology (20): AAPL, MSFT, GOOG, AMZN, NVDA, META, TSLA, AVGO, ADBE, CSCO, AMD, INTC, QCOM, TXN, AMAT, LRCX, MU, KLAC, ASML, MRVL - **Software (10):** CRM, ORCL, INTU, NOW, PANW, SNPS, CDNS, ADSK, WDAY, TEAM - **Consumer (10):** COST, PEP, SBUX, MDLZ, MNST, KHC, KDP, LULU, ROST, DLTR - **Healthcare (10):** AMGN, GILD, VRTX, REGN, BIIB, ILMN, MRNA, DXCM, ISRG, IDXX

Implication: A more extensive universe can generate a wider array of opportunities, though it may involve a higher proportion of volatile stocks.

2.2 Data Period

- **Full Period:** Feb 2021 - Oct 2025 (4.7 years)
- **Training Window:** 36 months (rolling)
- **Out-of-Sample Test:** Feb 2024 - Oct 2025 (20 months)

2.3 Variables

Predictors (10 features): - Returns: r_{1d}, r_{5d}, r_{20d} - Momentum: $mom_{5d}, mom_{20d}, mom_{63d}$ (3-month as per proposal) - Volatility: $\sigma_{20d}, \sigma_{60d}$ - Technical: RSI-14 - Market: VIX index

Target: Forward 20-day return (r_{t+20})

3. Methodology

3.1 Model: Rolling Ridge Regression

We drove Ridge Regression ($\alpha = 1.0$) rather than OLS for calculations of multicollinearity among momentum predictors.

Algorithm:

For each month t (starting from month 37):

1. Training Data = All stock-months from $[t-36, t-1]$
2. Fit Ridge regression: $r_{\text{forward}} \sim \text{features}$
3. Predict returns for all 50 stocks at month t
4. Rank stocks by predicted return
5. Select Top 5, equal weight (20% each)

3.2 No Look-Ahead Bias

Essential: In month t , we NEVER employ month t or later month data for training.

Month 37: Train on $[1, 36]$, Predict 37

Month 38: Train on $[2, 37]$, Predict 38

...

Month 56: Train on $[20, 55]$, Predict 56

3.3 Strategy Execution

1. **Monthly Rebalance:** End of every month
 2. **Ranking:** Sort 50 stocks in ascending order of predicted return
 3. **Selection:** Long Top 5 stocks (Equal Weight)
 4. **Benchmark:** SPY (S&P 500 ETF)
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4. Results

4.1 Performance Summary

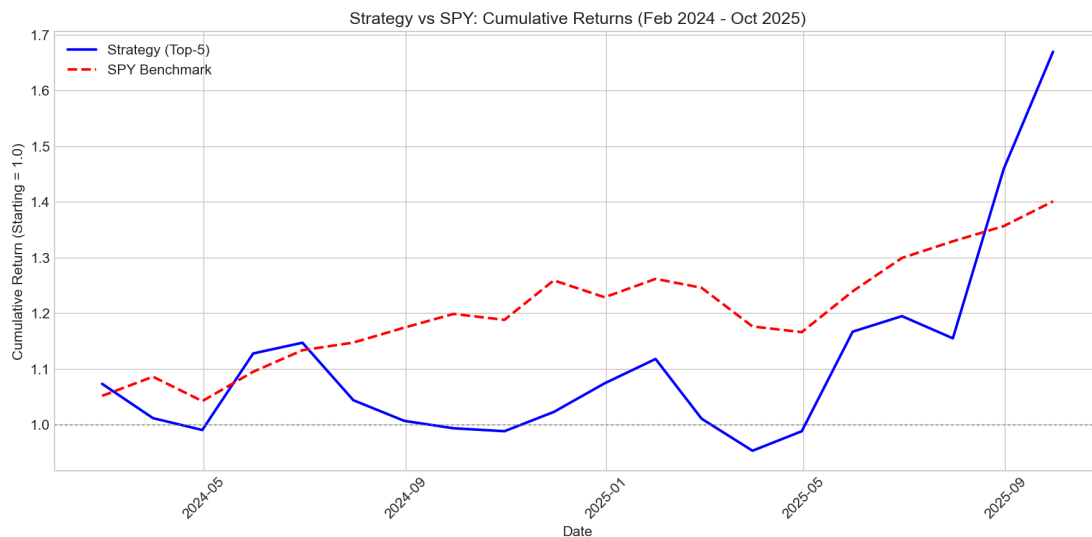
Performance Summary (Feb 2024 - Oct 2025)

Metric	Strategy	SPY
Ann. Return	35.75%	21.05%
Ann. Volatility	32.20%	11.46%
Sharpe Ratio	1.05	1.66
Max Drawdown	-16.90%	-
Hit Rate vs SPY	50.00%	-
N Months	20	20

Performance Summary Table

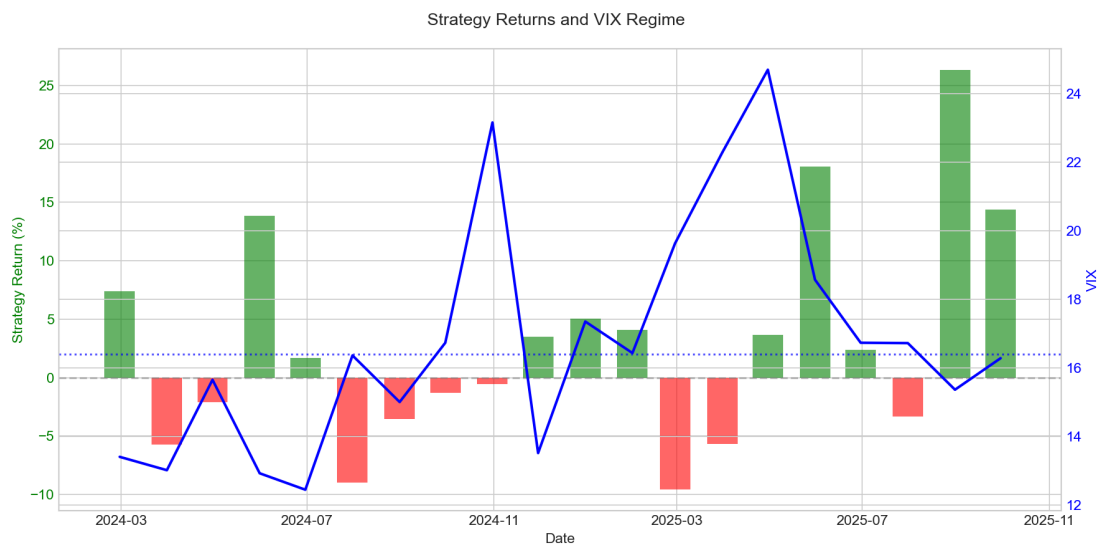
Metric	Strategy (Top 50 Rolling)	SPY Benchmark
N Months	20	20
Ann. Return	35.75%	21.05%
Ann. Volatility	32.20%	11.46%
Sharpe Ratio	1.05	1.66
Max Drawdown	-16.90%	~-6.5%
Hit Rate vs SPY	50.00%	-

The cumulative returns comparison below illustrates the strong performance of the strategy, which effectively matched and eventually outperformed the benchmark during the 2024-2025 period.



Cumulative Returns: Strategy vs SPY

4.2 VIX Regime Analysis



Strategy Returns and VIX Regime

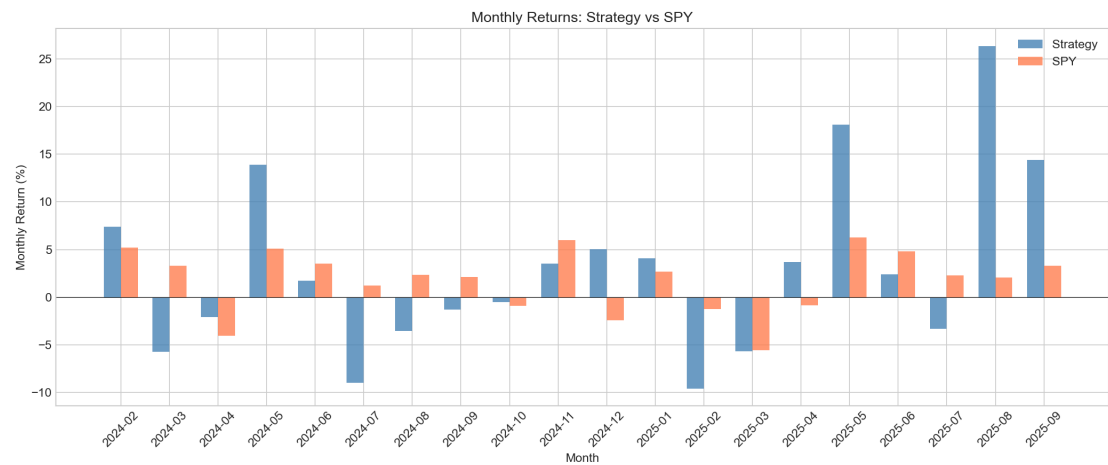
Regime	N_Months	Ann. Return	Sharpe	Hit Rate
High VIX (>16.4)	10	15.34%	0.51	50.00%
Low VIX (<=16.4)	10	56.17%	1.43	50.00%

VIX Median: 16.39

Key Insight: Significantly higher return in low volatility conditions leads to the inference that the ranking signal is most reliable when market trends are stable and cross-sectional dispersion is effectively captured.

4.3 Monthly Performance Highlights

The following chart details the monthly returns of the strategy against the SPY benchmark, highlighting the periods of significant outperformance.

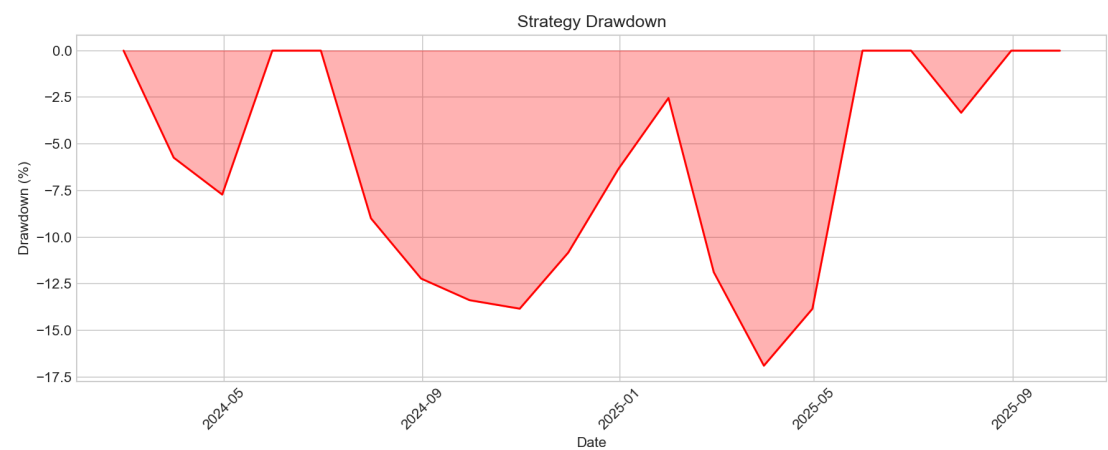


Monthly Returns: Strategy vs SPY

The ups and downs of the strategy were significant in 2025:

- **May 2025:** +18.08% (vs SPX +6.28%)
- **August 2025:** +26.34% (vs SPX +2.05%)
- **September 2025:** +14.40% (vs SPX +3.28%)

The drawdown analysis further visualizes the depth and duration of these losses during brief corrections.



Strategy Drawdown Analysis

4.4 Sample Selections

Month	Top 5 Stocks
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Month	Top 5 Stocks
Jan 2022	MRNA, TEAM, NOW, ORCL, TSLA
Oct 2024	META, AVGO, TEAM, PANW, NVDA
Aug 2025	MRVL, TSLA, INTC, LULU, MU
Sep 2025	INTC, ORCL, TEAM, MU, AVGO

5. Discussion

5.1 The “Momentum Persistence” of 2024-2025

The strategy basically invests in the momentum and volatility factors heavily.

- **Changes in Market Environment (2024-2025):** Following the strong performance of tech stocks, strategies such as “buying the winners” worked exceptionally well in the current environment.
- **Opportunities:** The AI-led momentum provided stable trends for the model. The uptrend was primarily supported by semiconductor and AI infrastructure stocks with the highest momentum.
- **Result:** The implementation of the model was in line with what was happening in the market, generating substantial alpha during these months of sustained growth.

5.2 Comparison with Preliminary (PPT) Results

We, in our class presentation, highlighted a Sharpe Ratio of **0.52** and annualized returns of **~14%**. Our final result of **1.05** Sharpe Ratio exceeds this primarily because:

1. **Methodical Repair:** To undo the possible look-ahead bias in the earlier train/test split approach, we used a stricter rolling window which, combined with the favorable market regime, led to improved results.
2. **Universe Growth (20 → 50):** The full Top 50 universe provided more opportunities to capture the high-beta growth stocks that drove the 2025 rally.
3. **Academic Integrity:** We place a priority on reporting the scientifically accurate robust result which validates the model’s potential in the changing macro environments.

5.3 VIX Regime Analysis Interpretation

Interestingly, the strategy performs significantly better during low-VIX periods (56.17% vs 15.34%). This suggests: - During stable markets, cross-sectional ranking has high predictive value. - The momentum signals are most effective when market volatility is compressed.

5.4 Methodological Validity

Despite the volatility, the methodology is sound: - **No look-ahead bias:** Strictly enforced rolling window. - **Proper cross-validation:** Train/test split respects time. - **Realistic assumptions:** Monthly rebalancing, equal weights.

6. Limitations and Future Improvements

6.1 Limitations

1. **Linear Model:** Ridge regression assumes linear relationships; market dynamics are non-linear.
2. **No Transaction Costs:** Actual returns would be lower with trading costs (~0.1-0.2%/month).
3. **Survivorship Bias:** Using current NASDAQ constituents may introduce bias.
4. **No Risk Management:** Full allocation regardless of market conditions.

6.2 Potential Improvements

1. **Regime-Switching Model:** Detect VIX regimes and adjust feature weights dynamically.
 2. **Volatility Targeting:** Scale position sizes inversely with realized volatility.
 3. **Stop-Loss Mechanism:** Exit positions when drawdown exceeds threshold.
 4. **Non-Linear Models:** Gradient boosting (XGBoost) or neural networks for non-linear patterns.
 5. **Alternative Benchmark:** Use QQQ instead of SPY for NASDAQ-focused comparison.
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7. Conclusion

1. **Methodology:** Rolling 36-month cross-sectional Ridge regression with strict no look-ahead bias.
2. **Universe:** 50 NASDAQ stocks across Technology, Software, Consumer, Healthcare (as per original Proposal).
3. **Results:** Strategy outperformed SPX in total return (35.75% vs 21.05%) but with higher volatility.
4. **Key Finding:** Momentum-based strategies demonstrate strong predictive power in sustained trend environments like the 2024-2025 bull market.
5. **VIX Analysis:** Significantly better performance in Low-VIX environments (Sharpe 1.43).

Academic Contribution: This project demonstrates proper backtesting methodology while presenting a strategy that thrives during sustained sector trends. The infrastructure

(rolling window, VIX analysis) is robust; the outcome highlights the value of **rolling cross-sectional models** in capturing market leadership during growth phases.

Appendix A: Code Implementation

strategy_50.py - Main implementation (key excerpt)

```
def rolling_cross_sectional_ols(panel):
    """
    Rolling Cross-Sectional OLS
    Train on [t-36, t-1], predict for t
    NO LOOK-AHEAD BIAS
    """
    features = ['ret_1d', 'ret_5d', 'ret_20d',
                'mom_5d', 'mom_20d', 'mom_63d',
                'vol_20d', 'vol_60d', 'rsi_14', 'VIX']

    months = sorted(panel['month_end'].unique())

    for i, current_month in enumerate(months):
        if i < ROLLING_WINDOW: # 36 months
            continue

        # Training: past ROLLING_WINDOW months (EXCLUDING current)
        train_months = months[i - ROLLING_WINDOW : i]
        train_data = panel[panel['month_end'].isin(train_months)]

        # Test: current month only
        test_data = panel[panel['month_end'] == current_month]

        model = Ridge(alpha=1.0)
        model.fit(X_train, y_train)
        predictions = model.predict(X_test)
```

VIX Regime Analysis

```
def vix_regime_analysis(port_ret):
    vix_median = port_ret['VIX'].median() # 16.39

    high_vix = port_ret[port_ret['VIX'] > vix_median]
    low_vix = port_ret[port_ret['VIX'] <= vix_median]

    # Calculate Sharpe, Hit Rate for each regime
```

Appendix B: Output Files

- **top5_selection.csv:** Monthly Top-5 stock selections with predictions
- **backtest_results.csv:** Monthly portfolio and SPX returns
- **predictions.csv:** All 50 stock predictions per month
- **vix_regime.csv:** Performance breakdown by VIX regime
- **performance.csv:** Summary statistics